



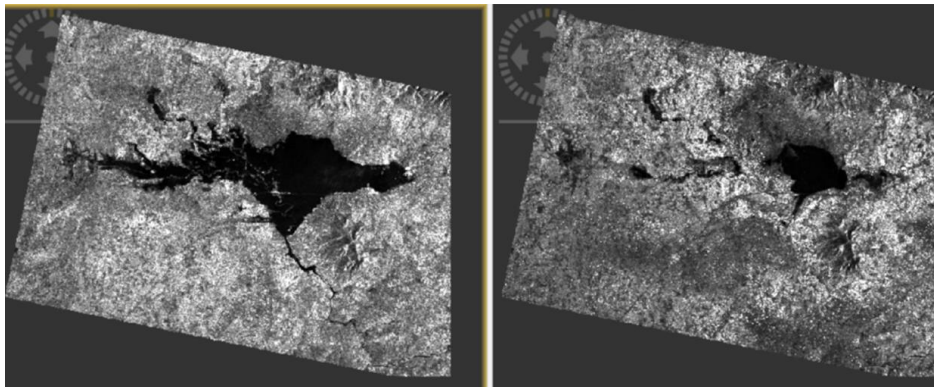
Bahir Dar University

Department of Geography and Environmental studies Masters in

Geo-Information Science

Advanced Remote Sensing Course

Practical Assignment on Using Sentinel-1 SAR imagery



Lab 1: SAR GRD for Flood mapping
Using SNAP and GEE.

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Date: May 15, 2026

1 Introduction and Background

Synthetic Aperture Radar (SAR) is an active remote sensing technology that generates its own microwave illumination. Unlike optical sensors such as Sentinel-2, SAR penetrates clouds, smoke, and haze while operating independently of sunlight. These capabilities make SAR indispensable for rapid flood mapping during severe weather events when immediate emergency response data is critical.

This lab introduces students to the core processing workflows of Sentinel-1 SAR Ground Range Detected (GRD) data using a recent real-world flood event in Ethiopia as a practical case study. Students will learn how to leverage SAR imagery to overcome traditional cloud-cover limitations in disaster management.

1.1 Use case: Problem and study area Context

Around August 7, 2025, heavy rainfall during the Ethiopian Kiremt rainy season caused severe floods across Oromia region—specifically within the Southwest Shewa and Awash River zones—and impacted parts of Addis Ababa. The disaster caused widespread displacement and infrastructure damage in rural and urban areas. This lab demonstrates how to detect, analyze, and map these affected regions by comparing "before" and "after" satellite imagery. We utilize two Copernicus Sentinel-1 SAR images: a "crisis" image captured during the flood on August 7, 2025 and a pre-flood image from July 26, 2025. Overlaying the crisis data onto the stable baseline allows us to isolate temporary floodwaters from permanent water bodies like rivers and lakes.

To achieve this, we will apply essential data pre-processing and transformations to the SAR GRD products to generate a binary flood map and calculate the total inundated surface area. This exercise is done using two known platforms-- SNAP and GEE; both accessible open-source software specialized platforms for SAR processing. The manual provides a comprehensive, step-by-step workflow from raw data ingestion to final flood area estimation, detailing the physical and geometric reasons behind each transformation step. While advanced analysis is beyond the immediate scope of this introductory lab, the workflow establishes the foundational technical skills required for subsequent advanced research and predictive modeling. We will conclude by discussing the potential implication of this lab exercise for final research project and suggesting advanced exercises that can be pursued building upon this lab.



Editor5 August 9, 2025

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Awash River flooding displaces thousands in West and Southwest Shewa

Flooding of the Awash River has displaced thousands of people across West and Southwest Shewa zones. Photo: Oromia Water and...

[Read More »](#)

Source: <https://addisstandard.com/tag/flooding/>

1.2 Lab Objectives and Activities

By the end of this lab exercise, students will be able to:

1. Acquire Sentinel-1 SAR Data: Search, select, and download Sentinel-1 SAR GRD products directly from the Copernicus portal.
2. Automate SAR Pre-processing: Utilize the SNAP Graph Builder and Batch Processing tools to automate core preprocessing steps, including Apply Orbit File, Radiometric Calibration, Range Doppler Terrain Correction, and Spatial and Polarization Subsetting.
3. Execute Bi-Temporal Change Detection: Implement a dual-image detection strategy to compare a "flood event" crisis image against a pre-flood "baseline" image.
4. Perform Statistical Thresholding: Apply log-transformation (dB conversion) and evaluate backscatter histogram distributions to determine optimal threshold values for separating water from land.
5. Generate Flood Maps: Use SNAP Band Math and change-detection ratios to isolate inundated areas and calculate total area.
6. Export Analysis-Ready Data: Export processed spatial layers into standardized, Analysis-Ready Data (ARD) formats like GeoTIFF and KMZ for downstream impact assessments.
7. Visualize Spatial Results: Overlay, style, and analyze the exported flood maps within Google Earth.
8. Perform the similar Flood mapping exercise using GEE(Google Earth Engine)
9. Present comparative evaluation, analysis and discussion on using SNAP vs. GEE for Flood mapping.

1.3 Datasets

The two datasets for this lab will be shared as **SAR_Lab_1** data in zipped folder, which contain Image 1: August (flood event, peak inundation) and Image 2: July (Pre-flood, stable ground).

1. [S1A_IW_GRDH_1SDV_20250807T031739_20250807T031804_060424_0782D1_A7C7.zip](#)
2. [S1A_IW_GRDH_1SDV_20250726T031739_20250726T031804_060249_077CBF_5A7F.zip](#)

However it is recommended to get the datasets from the source. Follow the precise steps and images below to download the required SAR datasets from the Copernicus Browser

1.4 Sentinel-1 Data Download Workflow:

Portal Access and Authentication

- Navigate to the web portal at <https://browser.dataspace.copernicus.eu/>
- Click the user icon in the top right corner.
- Select Login or complete a new registration profile

Define the Spatial Boundary (AOI)

The target area spans the Southwest Shewa and West Shewa zones along the Awash River.

- Use the map search bar to type Addis Ababa, Ethiopia to center your screen over the region.
- Zoom out slightly to see the western and southwestern river valleys.
- Click the Draw a polygon or Draw a rectangle tool on the right-side map toolbar.
- Sketch your bounding box as shown in the figures below.

Configure Search Criteria & Filters

- On the left-side panel, select the Search tab and apply these exact constraints:
- Data Source: Check the box next to Sentinel-1.
- Product Type: Choose L1 Ground Range Detected (GRD).
- Sensor Mode: Select Interferometric Wide (IW).

- Polarization: Set to Dual-polarization VV+VH.

Search and Download the Two Target Scenes

You must run two distinct temporal searches to grab the necessary baseline and crisis images.

Image 1: Inundation Event (Crisis Image)

- Change the date range filter from August 7, 2025 to August 9, 2025.
- Click Search to locate the target scene acquired on August 7, 2025.
- Click the Product Info icon to verify it fully covers your drawn AOI polygon.
- Click the Download Product button (the zipped file will be approximately 1.5 GB).

Image 2: Pre-Flood Baseline (Stable Image)

- Set the date range filter from July 24, 2025 to July 26, 2025.
- Click Search to find the scene acquired on July 26, 2025.
- Click the Product Info icon to verify that this file matches the exact same Path and Relative Orbit Number as the August 7 scene to ensure consistent radar geometry.
- Click the Download Product button to retrieve the zipped file.

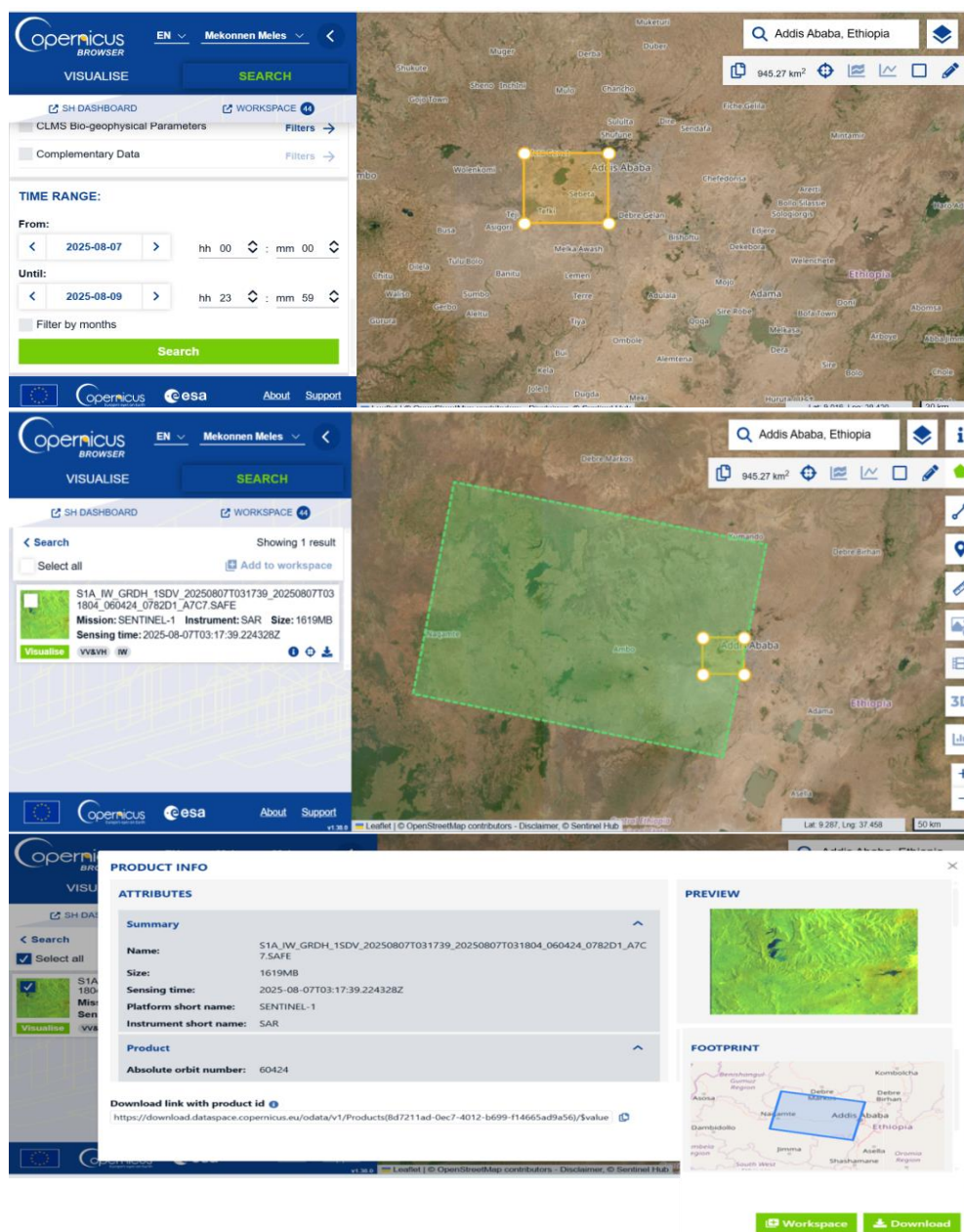


Figure 1: location of the data and Download steps

1.5 Software Requirements

- For this Lab, we use the Open source ESA SNAP 13 (Sentinel Application Platform). It can be downloaded from <https://step.esa.int/main/download/snap-download/>.

2 Data Import and Initial Product Exploration

After downloading you should have the two datasets in your data folder

1. S1A_IW_GRDH_1SDV_20250807T031739_20250807T031804_060424_0782D1_A7C7.zip
2. S1A_IW_GRDH_1SDV_20250726T031739_20250726T031804_060249_077CBF_5A7F.zip

Before beginning the processing chain, it is essential to understand the structure of the Sentinel-1 Level-1 GRD product which comes in a single .zip product. A single product contains multiple metadata files and measurement folders; however, SNAP simplifies this by treating the entire .zip folder as a single database.

- Load Products to SNAP and explore
- Launch SNAP. Use the Open Product button and navigate to your "SAR_Lab1_Data" folder. You do not need to unzip the original Sentinel-1 files; SNAP reads the .zip archive directly.
- Select both the Pre-flood (July 26) and Post-flood (August 7) products.
- In the Product Explorer, expand the folder structure: [1] S1A... > Bands.
- Double-click Intensity_VV for both products to open the image views.
- Click on Windows>select tile horizontally or vertically to see both images
- Explore product visually to identify specific AOI and distortions to be corrected.

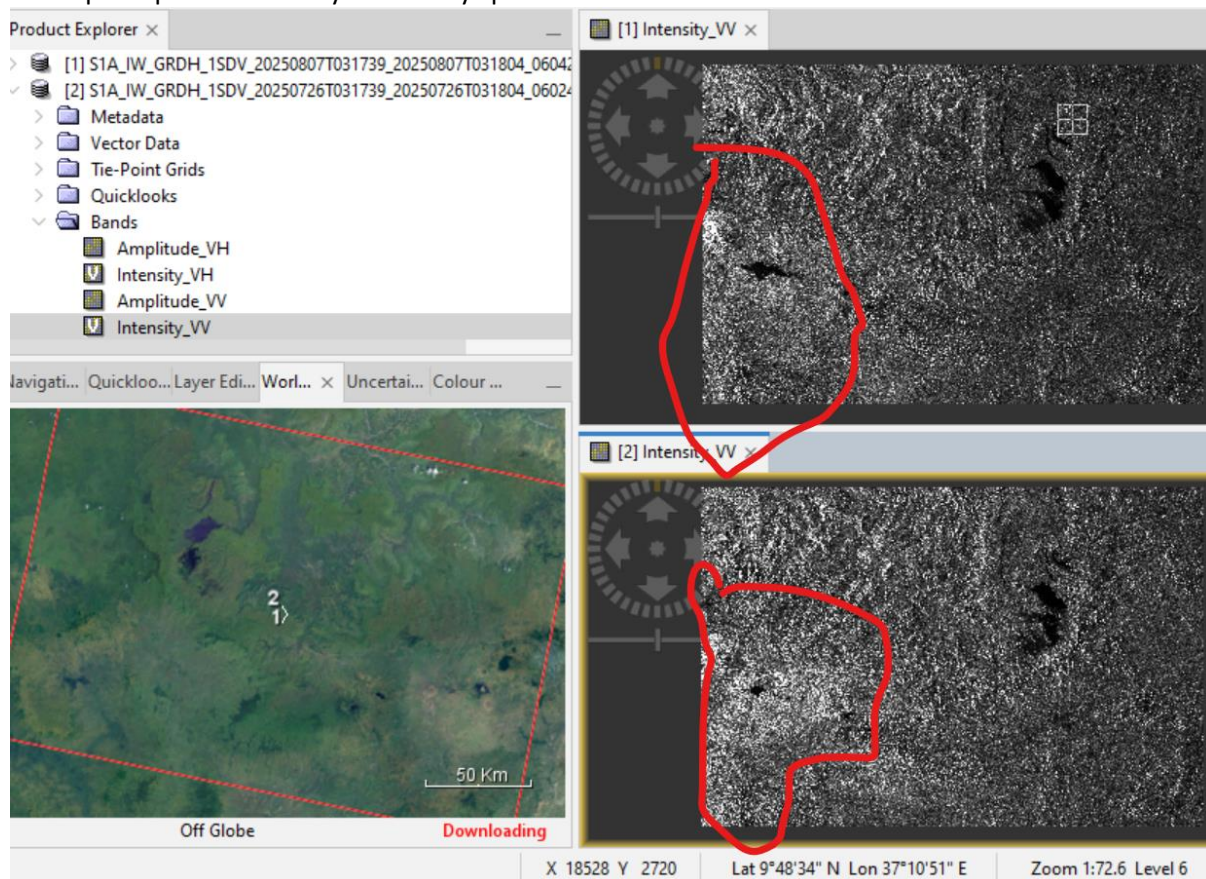


Figure 2: Overview of the two SAR GRD products in SNAP Product Explorer

Observe the images flood event (August) (Top) and Pre-flood (July 26) (bottom) and you will notice difference in water areas seen as black, and urban areas in white. You can also zoom in and check

the images to see if we need to subset. You may also notice the image appears disoriented, inverted or distorted; this is because the data is still in the satellite's radar geometry and has not yet been terrain-corrected. We will handle these in the pre-processing stage.

2.2 Explore AOI for Subsetting

The full Sentinel-1 IW GRD product spans roughly $110\text{km} \times 150\text{km}$, which can be significantly larger than our AOI. Subsetting reduces processing area, time and memory use. In this lab we are interested flood mapping focusing on areas around (Awash River) and parts of urban (Addis Ababa) (red circle in figure 2).

- Open one product in SNAP and display its Amplitude_VV band on screen.
- Zoom in to our AOI by including Awash River and Addis Ababa areas.
- Right-click on the image and select Spatial Subset from View to open the subset dialog window.
- The AOI to subset from the large image can be configured using the shp file provided.
- For now just explore and click Cancel to close the subset dialog window.

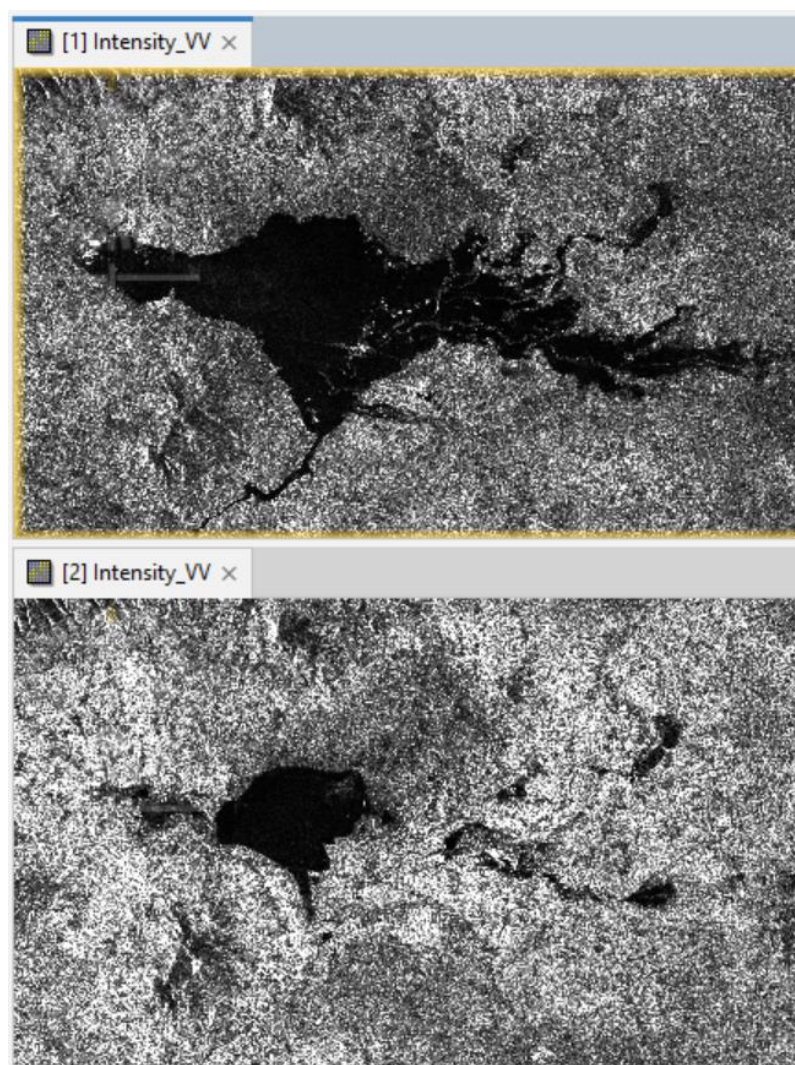


Figure 3: Our AOI (Area of Interest) includes Awash River and parts of Addis Ababa.

3 Pre-Processing:

3.1 Using the Automated processing Graph

Individual manual processing of SAR images is prone to human error and is inefficient for time-series analysis. We will use the Graph Builder to define a process model once and re-use it several times.

3.2 Building the Pre-processing Graph

- Go to Tools > Graph Builder. A default graph with "Read" and "Write" nodes will appear.
- Right-click in the white space of the graph window to add the following operators (nodes) in order:
- Radar > Apply Orbit File: To update the metadata with precise satellite orbital positions.
- Radar > Radiometric > Calibration: To convert raw intensity into scientifically meaningful Sigma0 values.
- Radar > Speckle Filtering > Single Product Speckle Filter: To reduce sandy noise while preserving edges.
- Radar > Geometric > Terrain Correction > Range-Doppler Terrain Correction: To project the data from radar geometry to a geographic coordinate system (WGS84) using a DEM.
- Raster > Geometric > Subset: To crop the data to the Addis Ababa / Awash region.
- Connect the nodes by dragging arrows from the right side of one node to the left of the next.

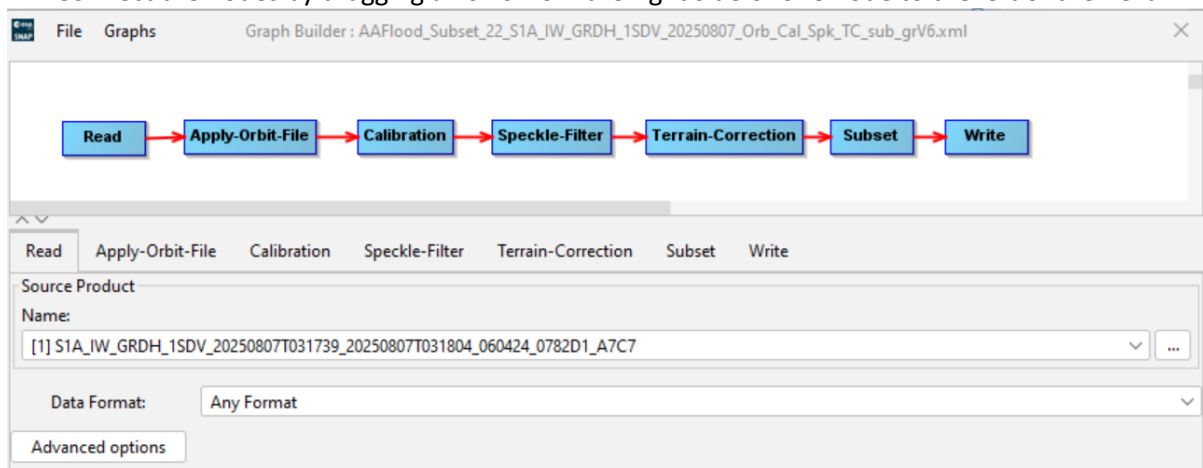


Figure 4: Graph for pre-processing Sentinel-1 SAR GRD data.

3.3 Configuring the Processing Graph Nodes

Each operator in the Graph Builder has specific parameters that must be set to ensure the output is correct and Analysis Ready Data (ARD). Click on each tab at the bottom of the Graph Builder window to configure the following:

3.4 Read Data:

- Under Source Product select the August image as shown above.
- Leave other parameters as the default.

3.5 Apply Orbit File

Sentinel-1 SAR metadata initially contains predicted orbits, updated precise orbits often become available after 28 days. To ensure high-accuracy and spatial alignment between our images, we must use the updated precise orbits.

- Go to Radar menu and select Apply Orbit File.
- Orbit State Vectors: Sentinel Precise (Auto Download)
- Polynomial degree:3
- Checkbox: Select "Do not fail if new orbit file is not found" (this allows the graph to proceed with Restituted orbits if Precise ones aren't available yet).

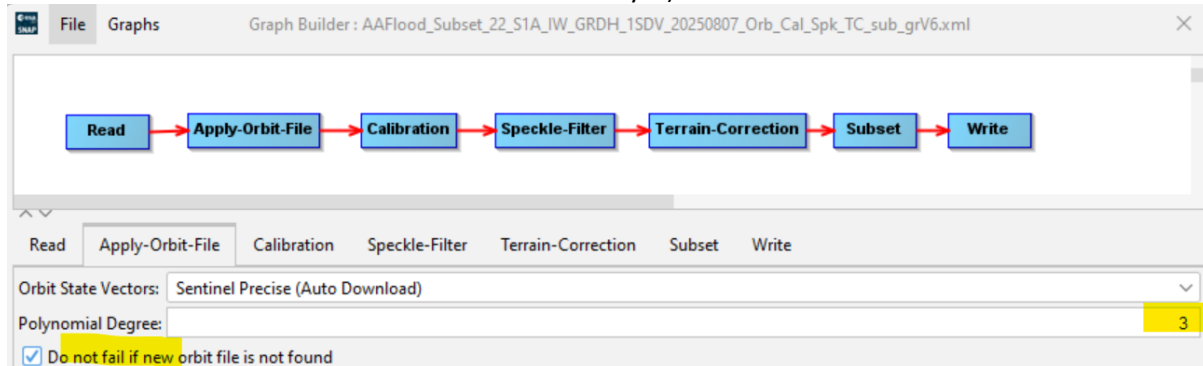


Figure 5: Apply Orbit File parameters

3.6 Radiometric Calibration

This is the most critical step for quantitative analysis. It converts the relative "brightness" of the radar raw data into the Sigma Nought backscatter coefficient.

- Output: Sigma0
- Note: Once this step is complete, your Intensity_VV/ Intensity_VH band will be renamed to Sigma0_VV/ Sigma0_VH. This value is measured in a linear scale which allows us to perform quantitative tasks easily.

3.7 Speckle Filtering

Speckle (salt and pepper / sandy texture) is a typical characteristic of SAR GRD data caused by random constructive and destructive interference from neighboring pixels backscatter.

Speckle filters can be applied to the data to reduce the amount of speckle at the cost of blurred features or reduced resolution. Several speckle filtering algorithms are available and the choice for a best filter often depends on the type of data, its spatial resolution, the degree of inherent speckle, and the application(user) intent.

For this lab, we will apply Lee Sigma speckle filtering as follows.

- Filter: Lee Sigma. This is a popular choice for urban areas as it smooths homogenous regions (like water) while preserving sharp edges (like buildings and roads).
- Window Size: 7x7. A larger window provides more smoothing but can blur small features.

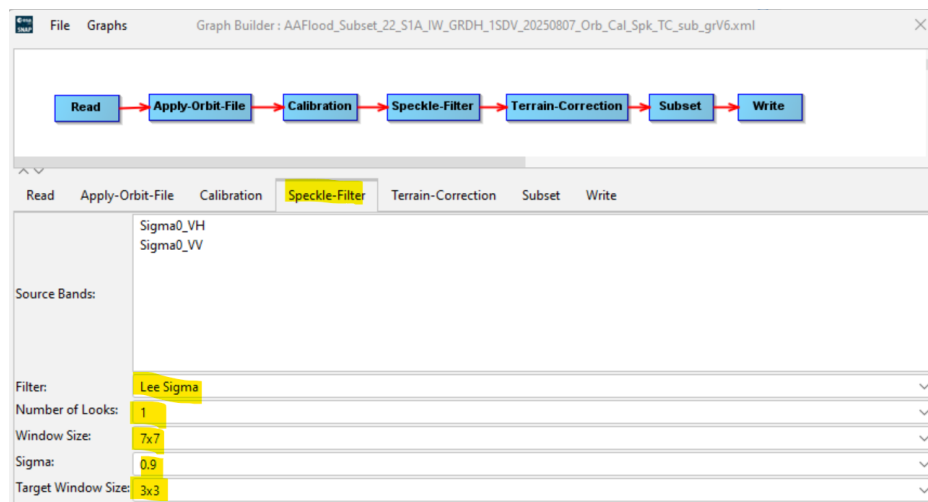


Figure 6: Speckle filtering parameters

Another popular algorithm we can use for our case is the Refined Lee filter; it averages the image while preserving edges. It has no parameters to set, while others require the definition of a kernel size and other parameters. The effect of different filters and their parameter configurations has to be explored by careful comparison to find the best solution for the respective case.

3.8 Range-Doppler Terrain Correction

Currently, the image is distorted due to the satellite's side-looking geometry effects like layover and foreshortening. Terrain correction addresses these issues and flattens the image onto a map.

- Select Terrain Correction Tab and set the following parameters as shown in the figure.
- Digital Elevation Model (DEM): SRTM 1Sec HGT (Auto Download). SNAP will automatically download the required elevation tiles for Ethiopia. Make sure you have internet connection.
- Map Projection: WGS84 (Lat/Lon) or WGS84 (DD).
- Pixel Spacing: 10.0m. This matches the native resolution of the Sentinel-1 IW mode.
- Leave other parameters as default.

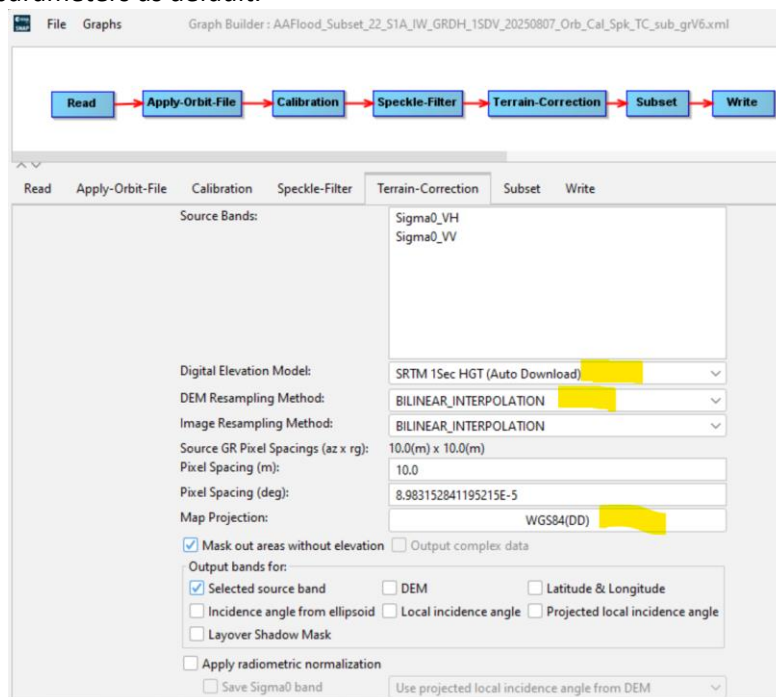


Figure 7: parameter settings for Terrain Correction

3.9 Subsetting AOI

Processing a full Sentinel-1 slice (swath) can be memory-intensive and time consuming. Creating a Spatial Subset focusing only on the Awash River and Addis Ababa areas is necessary to be efficient in subsequent processes. Subsetting also refers to choosing polarization of our interest. In this lab we will focus on VV as it provides the best contrast for detecting calm water surfaces.

For this lab we will import and use the aoI boundary shape file shared with other lab materials. SNAP will automatically extract the necessary dimensions shown in figure 8.

- Source Bands: Select Intensity_VV.
- Spatial Subset: Use the "Polygon" option button and click "Import from vector file" as shown in figure 8. Then find the aoI provided for this lab and click OK.

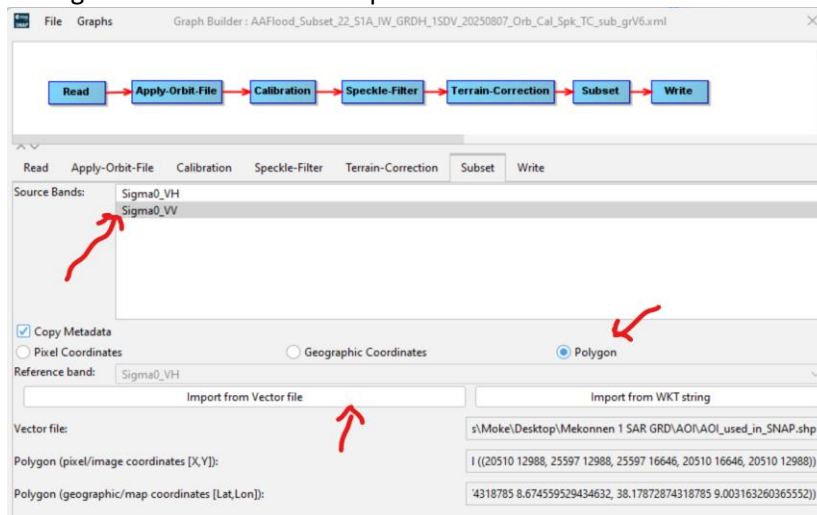


Figure 8: Spatial and polarization Subsetting using AOI

3.10 Write (save) product

- Select the directory to save the final output data.
- Leave the default name as it is.

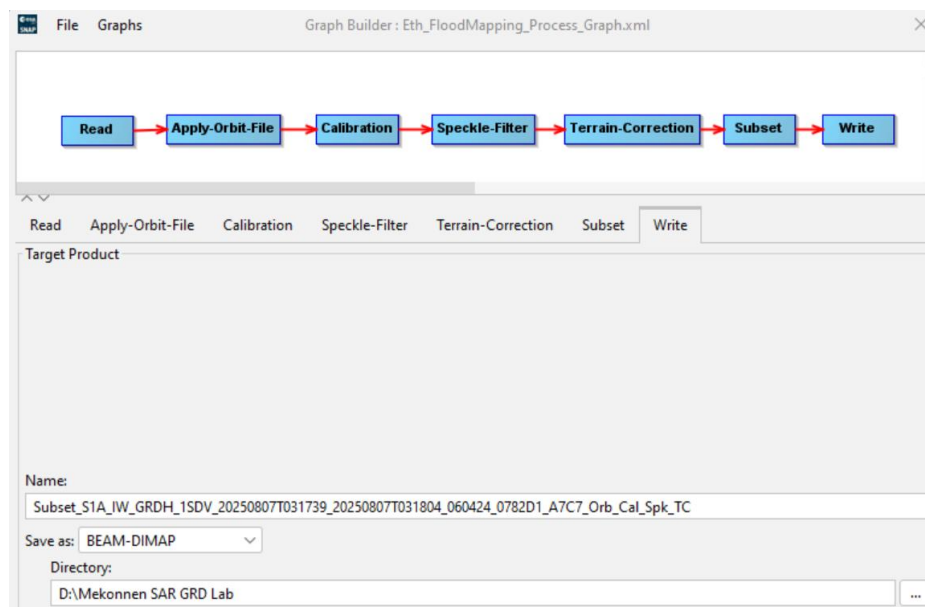


Figure 9: Write tab parameters

3.11 Save the graph for batch processing

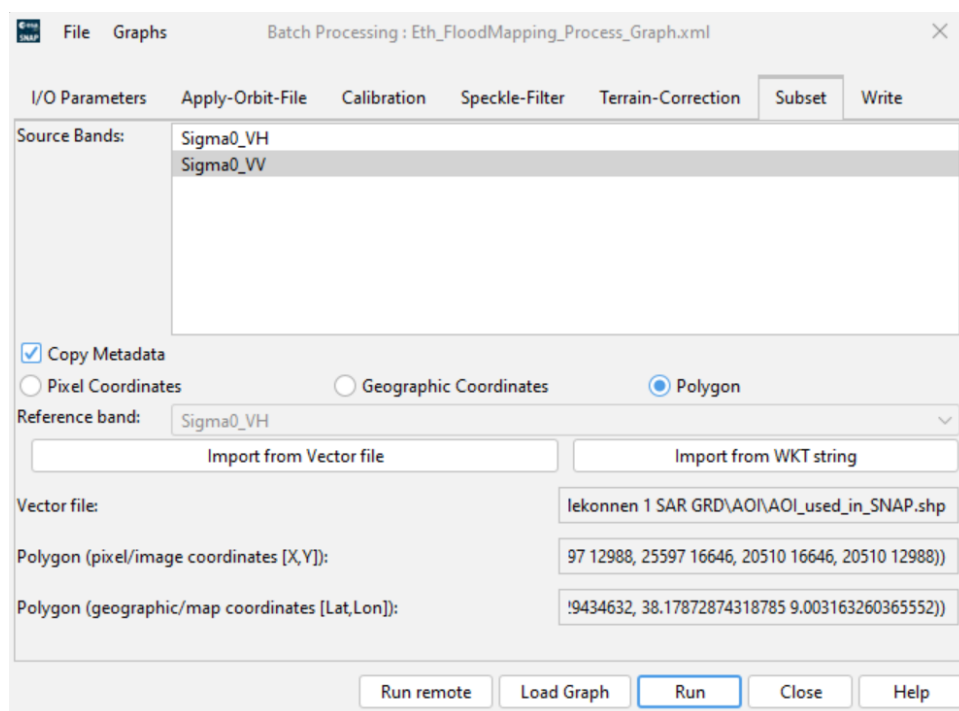
Save the graph for batch processing using File > Save Graph and provide appropriate name (e.g., Eth_FloodMapping_Process_Graph.xml).

3.12 Batch Processing

Now that the Graph (the instructions) is built, we do not run it manually for each image. Instead, we use Batch Processing to ensure both the July and August images are processed with the exact same mathematical parameters. Which is a requirement for multi-temporal (time series) analysis.

3.12.1 Configure Batch processing Tool

- Navigate to Tools > Batch Processing.
- I/O Parameters: Click "Add Opened" icon to bring in your two raw products from the Product Explorer.
- Load Graph: Click the "Load Graph" button at the bottom and select the XML file you saved.
- Tabs Check: You will notice new tabs appearing in the Batch window corresponding to your graph nodes. Quickly click through them to ensure your settings are still correct.
- Especially, double check the parameters in the Subset tab. To be sure Select Sigma_VV band and Import the aoi shape file again.
- Write Tab: Choose right directory again; and leave file name and format as default.
- Click Run. SNAP will now process both images back-to-back using the exact same parameters.
- A progress bar will appear at the bottom left. SNAP will start by downloading orbit files, downloading the SRTM DEM for Ethiopia, and performing the calculations. The overall batch processing(for binary image) may take from 5–25 minutes depending on your internet speed and computer capacity.



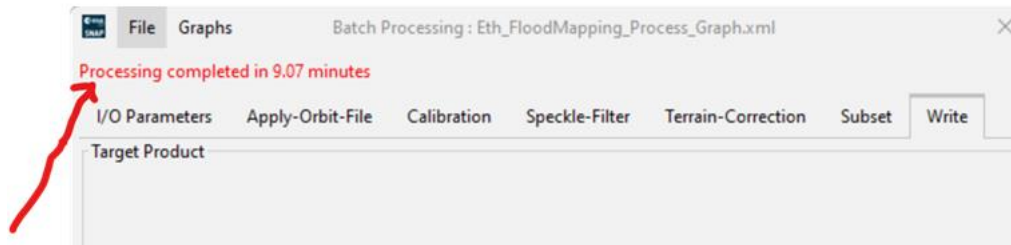


Figure 10: Batch Processing

3.13 Visual Inspection of batch pre-processing Output

Once the batch processing is complete, we will have two new .dim products in your output folder. SNAP will automatically load the two new products to product explorer. If not we have to load them ourselves.

- NOTE: The name of the two new products start with Subset and end with (...Orb_Cal_spk_TC), which indicates all the preprocessing steps applied.
- Open the Sigma0_VV band for the both images.
- Go to Windows menu and click Tile vertically or horizontally.

Observe the changes before and after pre-processing, like texture and orientation of terrains and Awash River (appearing black); the geometry is now corrected and flipped; aligns with a standard map.

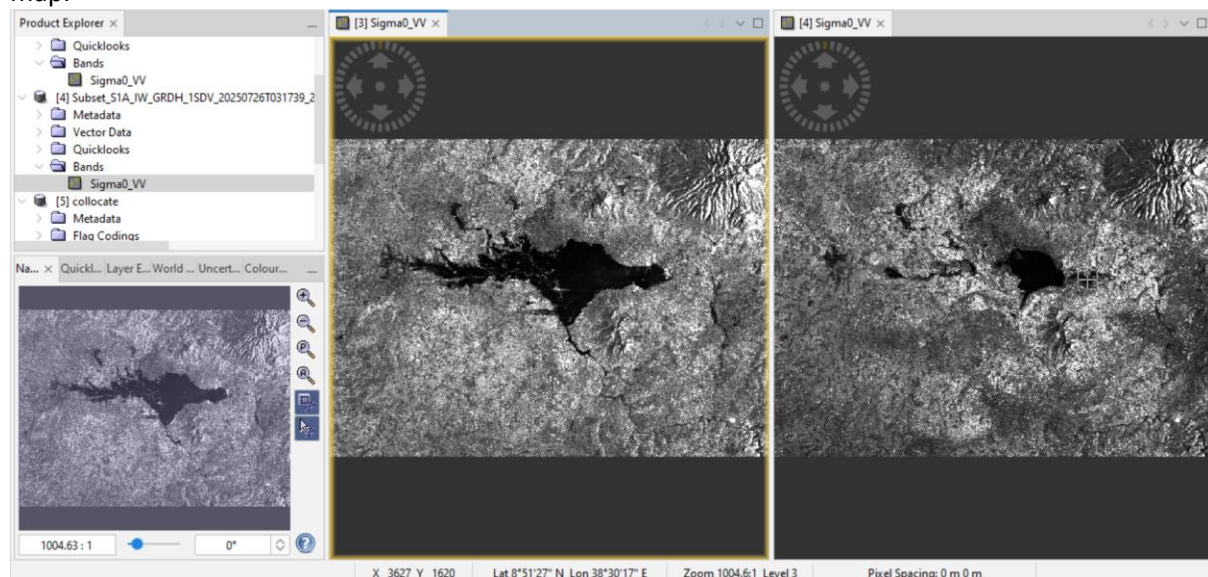


Figure 11: overview of images after pre-processing, highlighting the dark specular returns of the flooded areas along the Awash River

4 SAR analysis (post processing)

SAR GRD Multi-temporal Analysis

4.1 Image Collocation (Stacking)

Once the individual Sentinel-1 products have been pre-processed into Analysis Ready Data (ARD), the next phase of the workflow is Collocation. Because SAR images are acquired at different times, there may be slight sub-pixel misalignments despite terrain correction.

Collocation (stacking) is a process which creates a single stack from multiple bands by aligning every pixel. This allows us to perform a pixel-by-pixel comparison between the archive (July) and the disaster (August) images.

- Go to Raster → Geometric Operations → Collocation.
- In the Reference (Master) image selection tab, choose the subsetting August (main flood event) image.
- In the Secondary (Slave) image selection tab, choose the subsetting July (Pre-flood) image and click the include metadata check box.
- Name: Collocate
- Keep using output directory as before
- In the Rename tab, notice the prefixes (usually _M for Master and _S for Slave). This helps us tell the dates apart.
- Resampling : use Bilinear interpolation resampling
- Click Run. A new product named Collocate will appear in the Product Explorer.
- Expand the Bands folder and open both images,
- Windows → Tile vertically

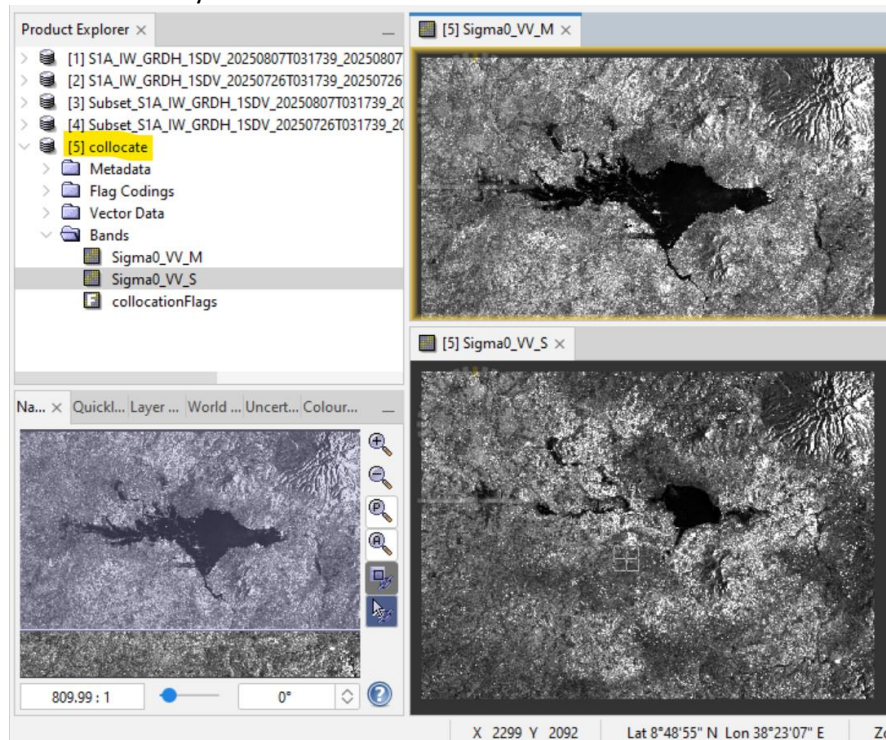
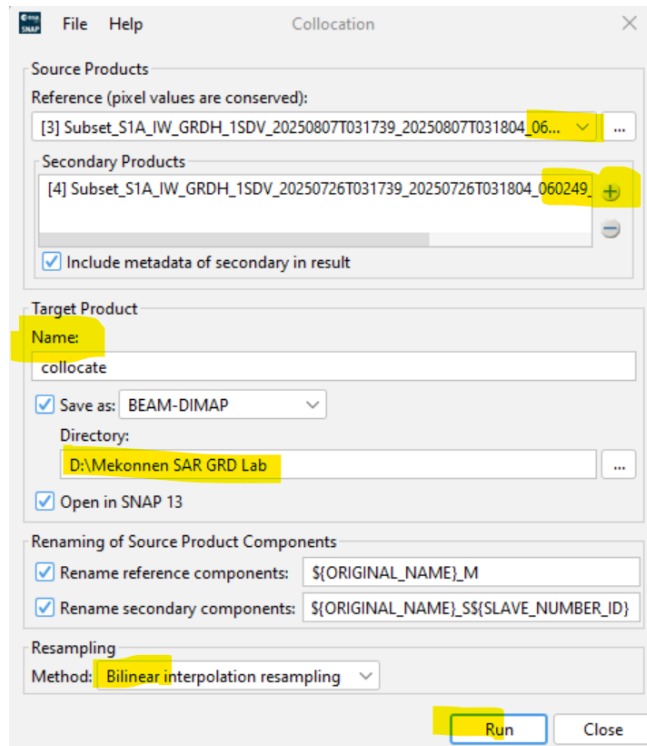


Figure 12: Collocated (stacked) view of both images

4.2 Linear to Logarithmic scaling

We can see that the images are dark and that from the histograms, in the 'Color Manipulation' window, most of the pixels have a small backscatter value and few of them have a high backscatter value. In this case it is better to transform the data from linear to logarithmic values or decibel (dB). In figure 13 below, we can see that the distribution of the pixel values in decibel is easier to understand and manipulate. This scaling also allows better visual contrast and Gaussian distribution for statistical Thresholding.

- Right-click on the August Sigma0_VV_M band.
- Select Linear to/from dB
- Repeat this for the July Sigma0_VV_S band.
- Two new virtual log10 scaled bands (Sigma0_VV_M_db and Sigma0_VV_S_db) are created. These are what we will use for the flood detection math.
- Open the 4 images(bands) under collocate and use the "Tile windows evenly", and the "Synchronize views and Cursor across multiple windows" tools to view and compare them as below.

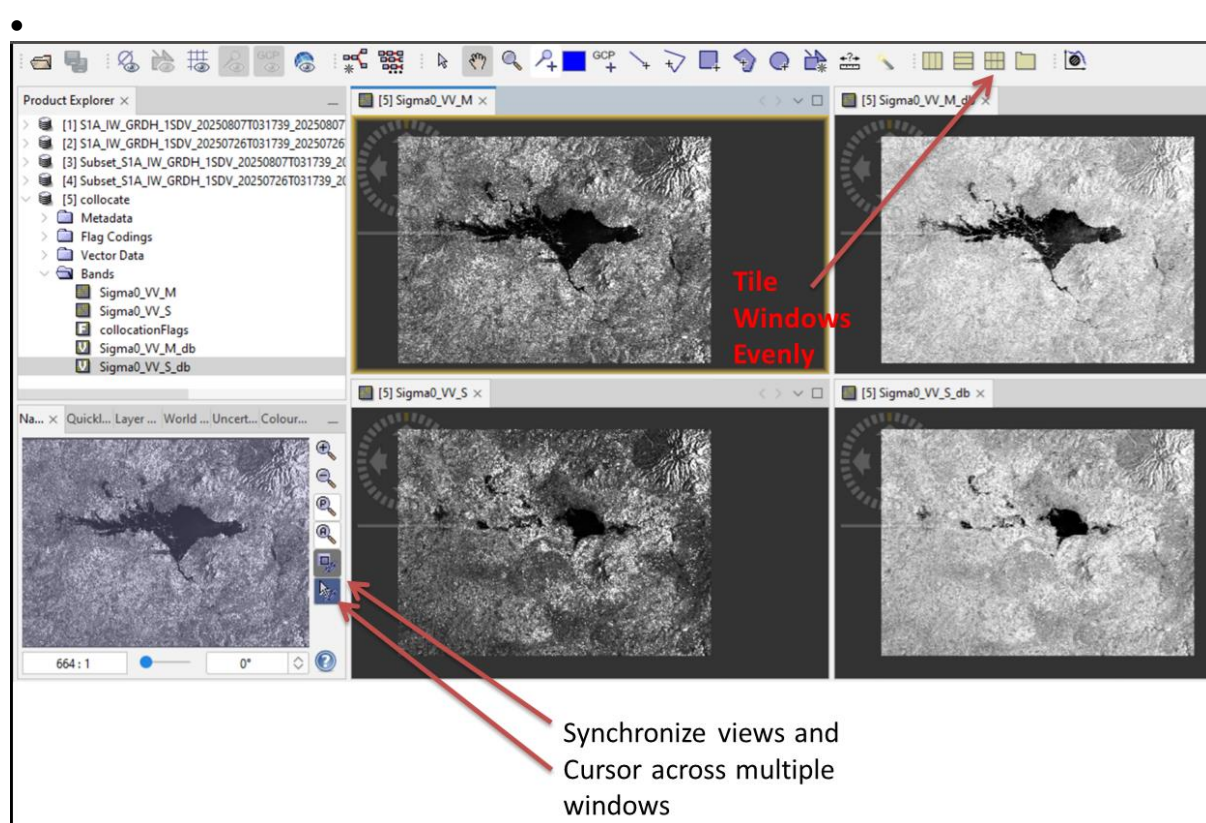


Figure 13: Linear sigma0(Left) and Log Scaled (dB)(Right side) view of the two VV images

4.3 Image Thresholding, Water Masking, and Raster Algebra

To isolate flood-induced inundation, a localized image thresholding technique was applied using SNAP's Band Maths engine. The primary objective of this binary classification process is to transform continuous radar backscatter coefficients (Sigma0) into a discrete thematic layer, assigning a value of one (1) to surface water and zero (0) to non-water matrices. To accurately isolate transient flood extents, this workflow requires a two-step approach: calibrating an operational backscatter threshold to extract the crisis footprint, and subsequently executing cross-temporal raster subtraction against the pre-crisis baseline archive to isolate newly flooded zones from permanent waterbodies.

4.3.1 Threshold Calibration and Bimodal Analysis

The separation of specular water bodies from rough terrestrial features is driven by the structural characteristics of the image histogram. Under ideal environmental conditions, a bi-temporal SAR scene over a partially flooded basin exhibits a distinct bimodal distribution. The first mode represents smooth, open water bodies characterized by low backscatter values owing to specular radar reflections. The secondary mode represents non-flooded terrestrial surfaces characterized by diffuse scattering and higher radar return intensity. The minimum variance point within the valley separating these two peaks defines the mathematical threshold limit. Pixels with backscatter coefficients falling strictly below this threshold are classified as flood zones, while pixels exceeding it are designated as unflooded land.

Histogram Analysis
To find the mathematically correct threshold for our region, we must analyze the histogram of one or both images as shown below. In a flooded image, the histogram often appears bimodal, showing two peaks: a low-intensity peak (water) and a high-intensity peak (land).

- Select the collocated product.
- Go to Analysis > Histogram.
- Click on the August (Sigma0_VV_M_db) band and click Refresh.
- Deselect the Log10 scaled bins checkbox as the data is already in decibel.
- Identify the dip depression or valley between the two peaks.
- Based on analysis below In this lab, we can use -12dB as our threshold,
- Do the same to analyze July (Sigma0_VV_S_db) image histogram and compare the intensity of water in both you can see that water content increased in august.

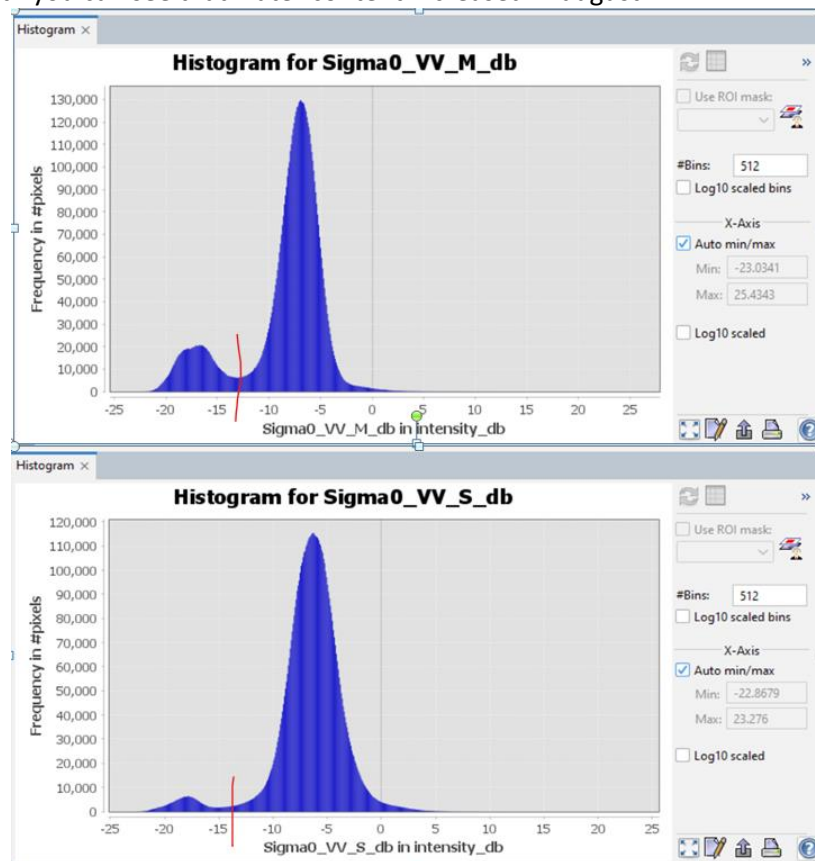


Figure 14: Histogram of the two images showing the bimodal distribution and the selected threshold line

4.4 Band Math and Creating the Water Masks

We will now use the Band Math tool to apply our threshold and generate three water masks.

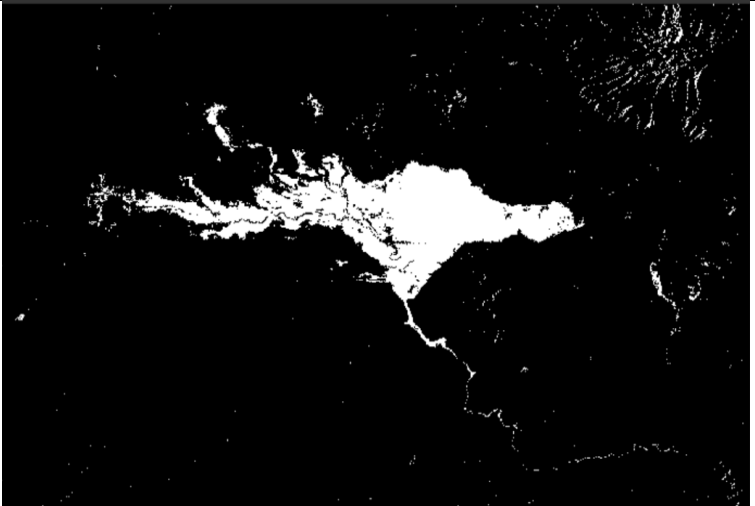
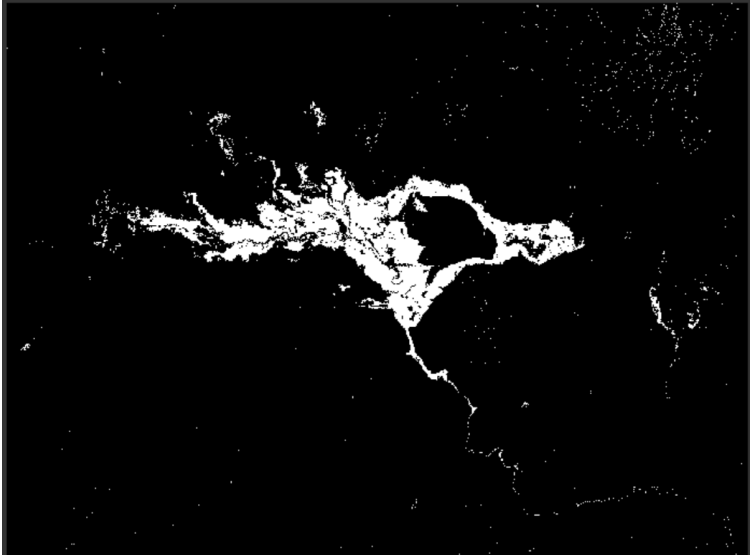
Band math	Out put
Find August water Mask: • Raster > Band Math. • Name: August_Water_Mask. • Uncheck "Virtual" (to ensure the mask is saved to disk). • Expression: $\text{if } \text{Sigma0_VV_M_db} < -12 \text{ then } 1 \text{ else } 0$ • Click ok • This expression assigns 1(white) to pixels that are water and 0(black) to non water areas.	
Find July water Mask: • Raster > Band Math. • Name: July_Water_Mask. • Uncheck "Virtual" (to ensure the mask is saved to disk). • Expression: $\text{if } \text{Sigma0_VV_S_db} < -12 \text{ then } 1 \text{ else } 0$ • This expression assigns 1(white) to pixels that are water and 0(black) to non water areas.	
Find newly flooded area • Perform Subtraction between two binary images. • Open Band Math again. • Name: flood_Map • Expression: $\text{August_Water_Mask} - \text{July_Water_Mask}$ • This expression identifies new water areas (i.e. Floods).Assigns white (1) to water pixels in august and Black (0) to others including water in July.	

Figure 15: Band math and visual comparison of the computed water masks: Top: July (Permanent Water); Middle: August (Flood + Permanent Water) and Bottom August flooded areas after the removal of the permanent water bodies.

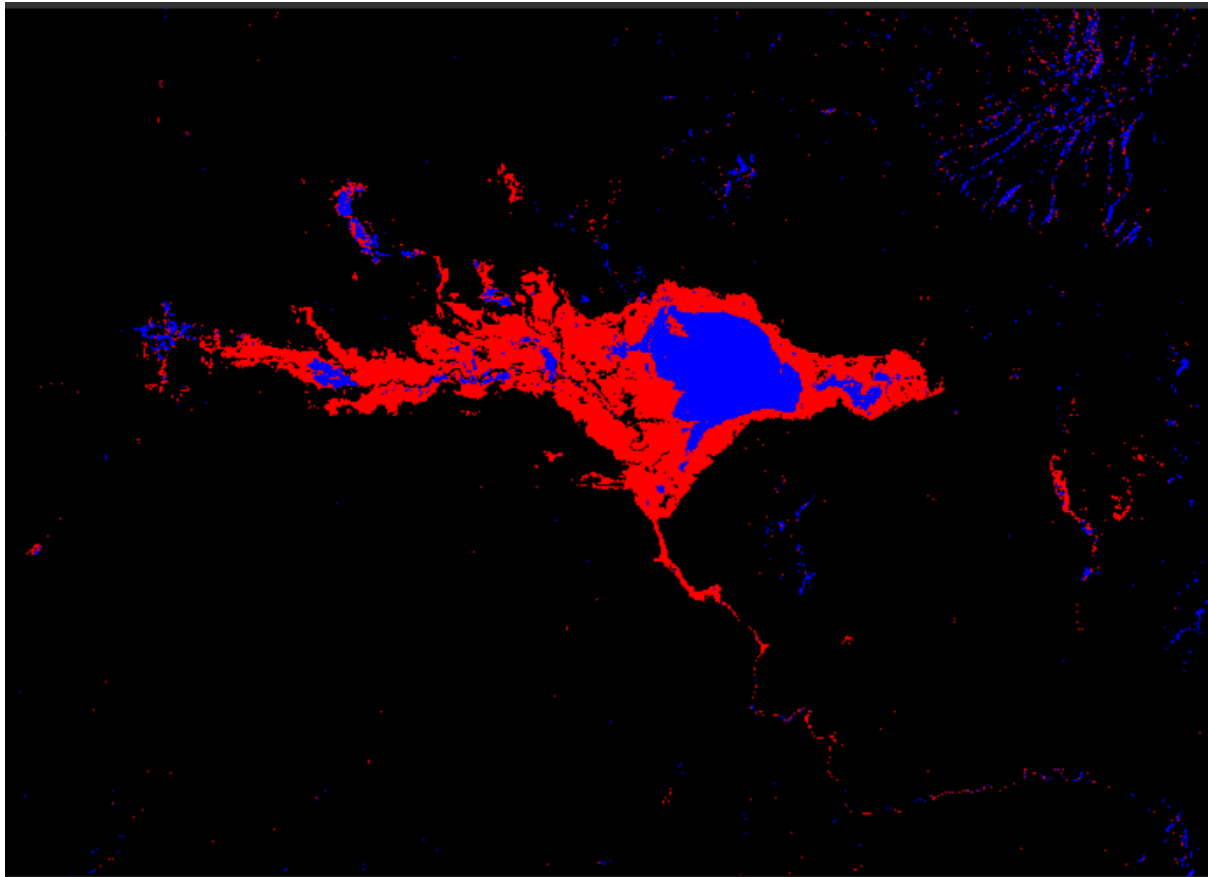


Figure 16: Flood (red) vs. permanent (blue) water view in SNAP Layer Editor

5 Estimating flood area

To compute the area covered by the flood we have to count the number of pixels in the flood_map binary image with value 1 and multiply that number for the area covered by a pixel (resolution). To count the number of pixels of the flooded areas we have to create a mask based on a logical band math expression as shown below.

- Select the stack(collocation) product,
- From the 'Mask Manager' window in right side, click on the 'F(x)' button.
- In the 'Logical Band Math Expression' popup window select 'Show bands' checkbox
- Under data source select the flood_Map and
- Write the expression ***flood_Map ==1*** as shown in the picture

Then we click on 'Ok'. We can change the name of the mask (new_mask_1) as e.g. ***flood_pixels_count*** in mask manager. This mask will select all the pixels whose value is 1 from the flood_Map binary image and can be used to calculate the flood area.

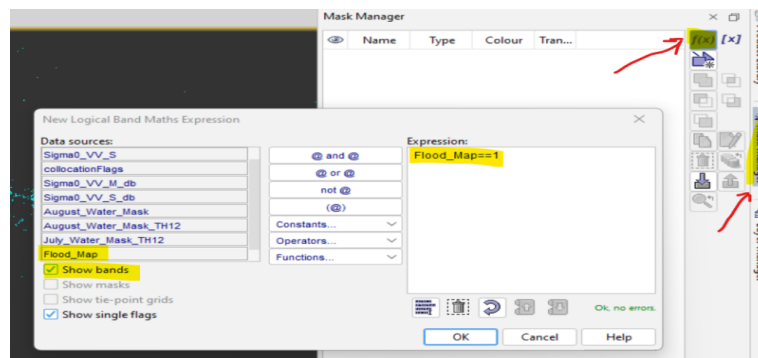


Figure 17: mask for counting water pixels

5.1 Calculate the flooded area,

- Go to mask manager and Check "flood_pixels_count" checkbox
- Raster -> Masks -> Mask Area (this tool starts computing the flooded area.)
- The result is shown in the 'Compute Mask Area' popup window as shown below.
- **The size of the flooded area is 85.856 Km².** This value can be somewhat different whenever we create new binary images.

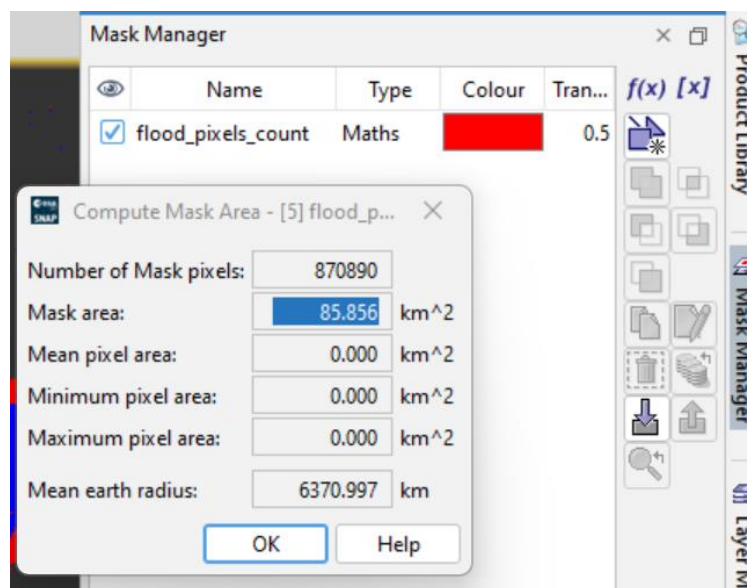


Figure 18 Flood area calculations

5.2 Suggestions for further analyses

The flood map generated in this lab represents typical application of SAR GRD imagery. However, for a meaningful environmental or socioeconomic impact study, we must combine these results with other thematic datasets and techniques.

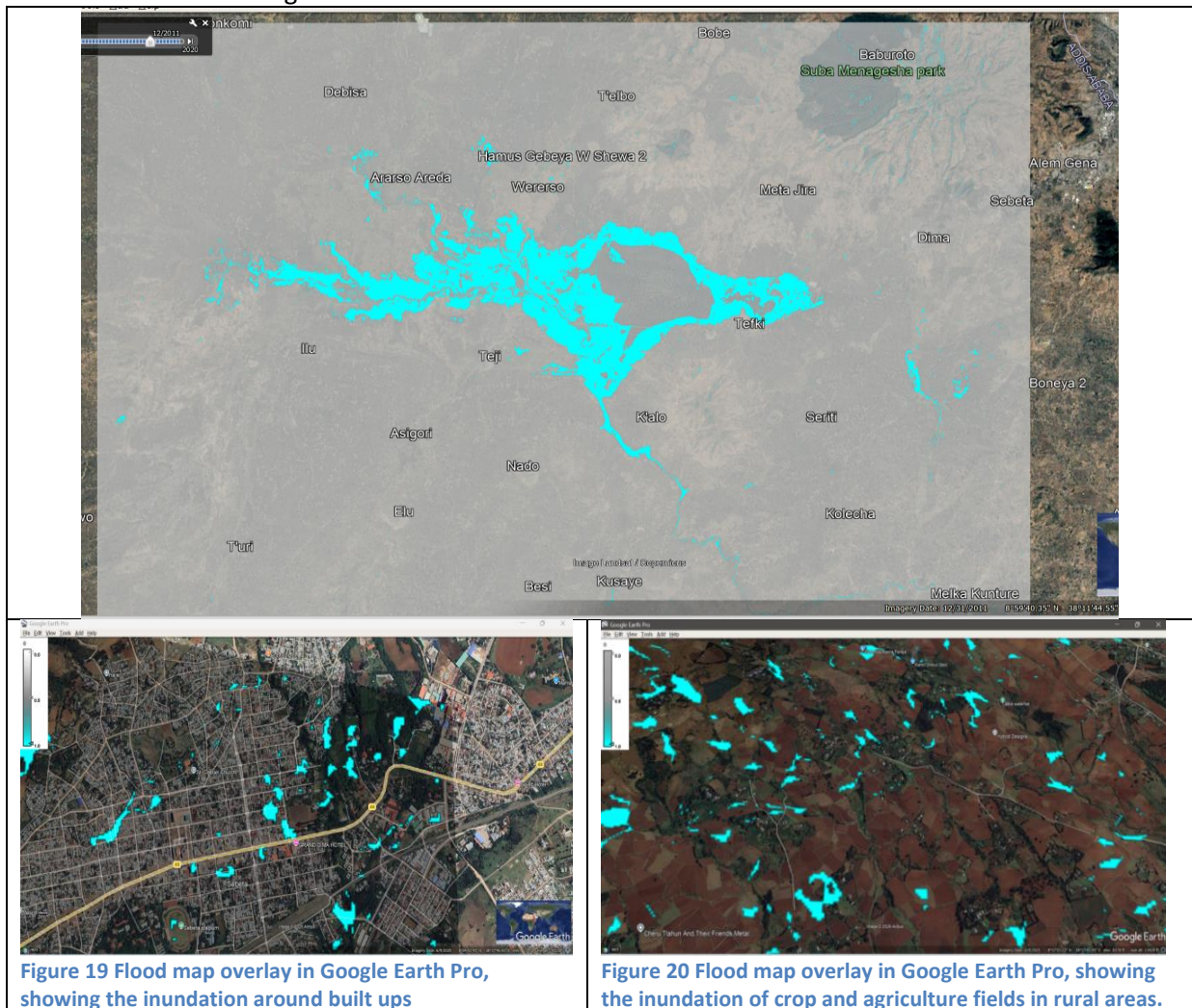
A suggestion on how to buildup on this lab exercise and do more analysis is given below.

- **LULC Overlay (Vulnerability Analysis):** By intersecting the flood mask with a Land Use/Land Cover (LULC) map, we can determine exactly how many hectares of Teff or maize were destroyed vs. how many urban structures were affected.
- **Population Exposure:** Integrating the flood extent with WorldPop or Gridded Population of the World (GPW) data to estimate the number of displaced persons in the West Shewa zone.

- Flood Frequency Mapping: If you have processed 5 years of SAR data, you can identify "Hotspots"—areas that flood every rainy season—to inform long-term rural/urban resilience planning.
- Hydrological Validation: Comparing the radar-derived flood extent with River Gauge Data from the Ministry of Water and Energy to correlate backscatter changes with water level height.

5.3 Exporting results

For further analysis, the results can also be exported into different size and formats compatible with Google Earth, QGIS, or ArcGIS. Navigate to File > Export > select format. For this lab we export KMZ for visualization in Google Earth.



Part 2: SAR processing with Google Earth Engine (GEE)

6 GEE work flow

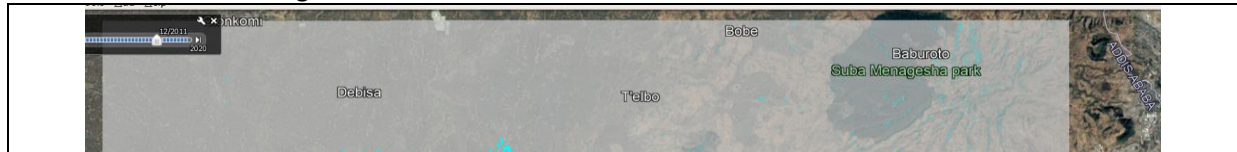
To understand SAR processing better and to validate the reliability and spatial consistency of spaceborne Synthetic Aperture Radar (SAR) flood mapping, we can perform identical bi-

BDU-Advanced remote sensing Lab: SAR GRD Processing By: Mekonnen G.

- Flood Frequency Mapping: If you have processed 5 years of SAR data, you can identify "Hotspots"—areas that flood every rainy season—to inform long-term rural/urban resilience planning.
- Hydrological Validation: Comparing the radar-derived flood extent with River Gauge Data from the Ministry of Water and Energy to correlate backscatter changes with water level height.

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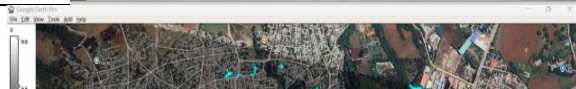




Figure 19 Flood map overlay in Google Earth Pro, showing the inundation around built ups

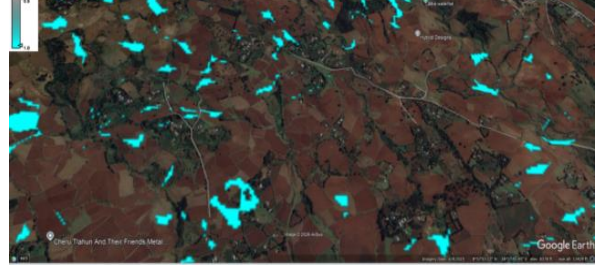


Figure 20 Flood map overlay in Google Earth Pro, showing the inundation of crop and agriculture fields in rural areas.

Part 2: SAR processing with Google Earth Engine (GEE)

6 GEE work flow

To understand SAR processing better and to validate the reliability and spatial consistency of spaceborne Synthetic Aperture Radar (SAR) flood mapping, we can perform identical bi-

temporal change detection using GEE .The following Google Earth Engine (GEE) script provides the bi-temporal flood mapping workflow as outlined in the SNAP.

It uses the Sentinel-1 GRD collection, applies a Thresholding strategy on the Awash River study area, creates a binary flood mask, calculates the flooded area in hectares, and prepares the layers for visualization. After the script comparative analysis of the two platforms is presented.

```
/**
 *
 *=====
 * BAHIR DAR UNIVERSITY - DEPARTMENT OF GEOGRAPHY & ENVIRONMENTAL STUDIES
 * Course: Advanced Remote Sensing | Student: Mekonnen Gidey (ID:
BDU1806929)
 * Project: Bi-Temporal Flood Mapping via Sentinel-1 SAR (Awash Floodplain)
 *
 *=====
 */

// 1. GEOGRAPHIC CONSTRAINTS
var aoi = table.geometry(); // please import the provided
AOI_used_in_SNAP_Polygon
// or use aoi below
//var aoi =
ee.Geometry.Polygon([[38.60,8.70],[38.60,8.92],[38.21,8.92],[38.21,8.70]]],
null, false);
Map.centerObject(aoi, 12);
Map.setOptions('SATELLITE');

// 2. SENTINEL-1 INGESTION & CORE FILTERING
var s1Collection = ee.ImageCollection('COPERNICUS/S1_GRD')
.filterBounds(aoi)
.filter(ee.Filter.eq('instrumentMode', 'IW'))
.filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
.filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VH'));

// 3. BI-TEMPORAL TIMEFRAMES & ORBIT TRACK ALIGNMENT
var preFloodColl = s1Collection.filterDate('2025-07-24', '2025-07-27');
var postFloodColl = s1Collection.filterDate('2025-08-07', '2025-08-09');

// Enforce identical track angles using the primary crisis image metadata
var firstPostImage = ee.Image(postFloodColl.first());
var matchingOrbit = firstPostImage.get('relativeOrbitNumber_start');
var matchingPass = firstPostImage.get('orbitProperties_pass');

var orbitFilter = ee.Filter.and(
  ee.Filter.eq('relativeOrbitNumber_start', matchingOrbit),
  ee.Filter.eq('orbitProperties_pass', matchingPass)
);

// Generate spatial mosaics clipped to the study area (VV Band Only)
var preVV_raw =
preFloodColl.filter(orbitFilter).mosaic().clip(aoi).select('VV');
var postVV_raw =
postFloodColl.filter(orbitFilter).mosaic().clip(aoi).select('VV');
```



```
// 4. ADAPTIVE LOCAL STATISTICS SPECKLE FILTER (Refined Reflected Code)
var applySpeckleFilter = function(image) {
  var linear = ee.Image(10).pow(image.divide(10)); // DB to linear
conversion
  var kernel = ee.Kernel.square(3, 'pixels'); // try 1 for 3x3; 3 for 7x7;
moving window

  var localMean      = linear.reduceNeighborhood(ee.Reducer.mean(), kernel);
  var localVariance = linear.reduceNeighborhood(ee.Reducer.variance(),
kernel);
  var noiseVariance = 0.25; // selected to match actual Sentinel-1
multilooking and speckle characteristics.

  // CALIBRATION ADJUSTMENT: test by increasing from 0.04 to 1
//var noiseVariance = 0.04; //resulted 112.3345852761005kmsq
// var noiseVariance = 0.25; // resulted 87.09379531558267
// var noiseVariance = 0.5; // resulted 85.24767501784169
// var noiseVariance = 1; //resulted 84.44188657308364

  // Higher values (0.25 - 1.0) increase smoothing resulting low area
  // Compute weight matrices (Lee adaptive scalar) to
  // increase smoothing of background noise while preserving sharp
boundaries.
  var weight =
localVariance.subtract(localMean.pow(2).multiply(noiseVariance)).divide(localVariance);
  weight      = weight.clamp(0, 1);

  var filteredLinear =
localMean.add(weight.multiply(linear.subtract(localMean)));
  return filteredLinear.log10().multiply(10); // Return to logarithmic
scale (dB)
};

var preVV = applySpeckleFilter(preVV_raw);
var postVV = applySpeckleFilter(postVV_raw);

// 5. BINARY THRESHOLDING & CONCURRENT SURFACE MASKING
var waterThreshold = -12; // Backscatter physical threshold cutoff for
specular water reflections
var changeRatio    = postVV.subtract(preVV);

var preWater = preVV.lt(waterThreshold); // Baseline water footprint
var postWater = postVV.lt(waterThreshold); // Post-crisis water footprint

// Isolate transient emergency waterbodies [Post-Water AND NOT Pre-Water]
var floodMask = postWater.and(preWater.not());
var floodMapOnly = floodMask.updateMask(floodMask); // Suppress dry
background matrices

// 6. PLANAR CARTESIAN METRIC AREA ESTIMATION
var floodAreaImage = floodMapOnly.multiply(ee.Image.pixelArea());

var areaStats = floodAreaImage.reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: aoi,
```

```
    scale: 10,                // Sentinel-1 native spatial resolution tracking
    crs: 'EPSG:32637',        // Enforce local UTM Zone 37N Planar Cartesian
    projection
    maxPixels: 1e9
  });

// Dynamic array index extraction bypassing structural dictionary band name
shifts
var floodAreaM2 = ee.Number(areaStats.values().get(0));
var floodAreaKM2 = floodAreaM2.divide(1e6);

// Console Diagnostic Metrics Output Logging
print('=====');
print('      BDU SAR LAB METRIC PROCESSING RESULTS      ');
print('=====');
print('Calculated Inundated Flood Area (Square Kilometers):',
floodAreaKM2);
print('=====');

// 7. CARTO-RENDERING & MAP DISPLAY LAYERS
var sarVis = {min: -25, max: 0};
Map.addLayer(preVV, sarVis, '1. Pre-Flood Baseline (July 2025) [dB]');
Map.addLayer(postVV, sarVis, '2. Flood Event Crisis (August 2025) [dB]');
Map.addLayer(changeRatio, {min: -10, max: 10}, '3. Change Detection Ratio
(Post minus Pre)', false);
Map.addLayer(preWater.updateMask(preWater), {palette: ['#0000FF']}, '4.
Permanent Water Bodies', false);
Map.addLayer(floodMapOnly, {palette: ['#FF0000']}, '5. Final Binary Flood
Extent Map');

// 8. CLOUD SYSTEM RASTER EXPORT
Export.image.toDrive({
  image: floodMapOnly,
  description: 'Awash_River_Flood_Mask_2025_Final_Mekonnen',
  scale: 10,
  crs: 'EPSG:32637', // Preserves strict local GIS vector alignment
  compatibility
  region: aoi,
  fileFormat: 'GeoTIFF',
  maxPixels: 1e9
});
```

6.1 GEE Map Outputs

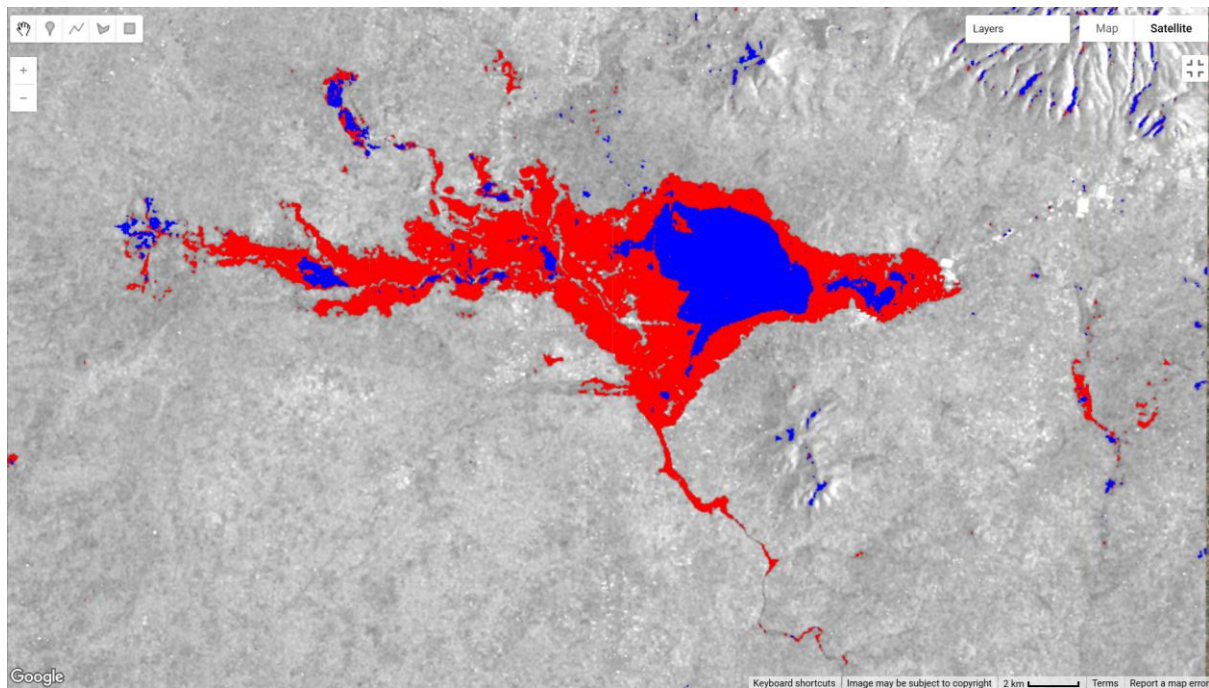


Figure 21: Flood (red) vs. permanent (blue) water view from GEE Layers

6.2 GEE Console Outputs.

- Pre-flood scenes found:1
- Post-flood scenes found:1
- **Calculated Inundated Flood Area (Sq. Kilometers): 87.09379531558267 kmsq**

7 Results and Discussion

7.1 Quantitative Flood Area Analysis and Platform Cross-Validation

The implementation of the SAR-based flood mapping workflow yielded highly comparable results across both platform architectures. Using an empirical threshold of -12 dB applied to the co-polarized vertical-transmit, vertical-receive (VV) channel, both engines successfully isolated the temporary inundation footprint along the Awash River plain and parts of Addis Ababa during the August 2025 Kiremt rainy season:

- Google Earth Engine (GEE) Output: The programmatic script calculated an estimated inundation area of **87.09 square kilometers (result found when replicating SNAP speckle filtering configurations)**.
- ESA SNAP Desktop Output: The final binary classified flood map generated an estimated inundation area of **85.86 square kilometers** based on the Mask Area tool.
- Cross-Platform Discrepancy: There is a minor cross-platform **discrepancy of approximately 1.23 square kilometers**.

Despite this minor variance, we can assume the SNAP workflow can be replicated in GEE. Both platforms effectively identified the primary flood boundaries, confirming that the physical behavior of calm open water (which causes specular scattering and low backscatter values) remains highly detectable and consistent across different software suites.

7.2 Workflow alignment and Sensitivity Testing (Optimization)

The initial cloud-based replication yielded an overestimation of 112.33 km² may be due to speckle filter weighting parameters. Investigation of work flow alignment was done to ensure congruency and address discrepancies. The process identified speckle filtering as primary source of discrepancy. Calibrating the local noise variance and using adaptive weighting to account for preserving edges effectively replicated the Lee Sigma filtering used in SNAP, bringing the final GEE calculation into cross-platform alignment with the desktop SNAP assessment framework.

Table 1: Platform Structural Workflow Comparison

To Step / Parameters Used	SNAP (Desktop Workflow)	GEE (Cloud Script)
Data Ingestion & Imagery Dates	Product Type: Sentinel-1 Level-1 GRD raw scenes. AcquisitionDates: Pre-flood baseline: July 26, 2025; Flood crisis event: August 7, 2025. Source Data: Copernicus Open Access Hub.	Product Type: ImageCollection('COPERNICUS/S1_GRD'). Acquisition Dates: Temporally filtered to match the explicit single-day timestamps: July 26, 2025 (Pre) and August 7, 2025 (Post).
Orbit Alignment & Pass Matching	Track Geometry: Manually cross-verified file properties to guarantee matching relative orbit paths.	Track Geometry: Programmatically enforced via metadata filters: matching relativeOrbitNumber_start and orbitProperties_pass.
Spatial Subset / AOI Ingestion	Vector Input: Esri Shapefile (.shp) polygon asset imported directly. Geometric Nature: Planar Cartesian clipping within the designated local coordinate framework.	Vector Input: Exact same Esri Shapefile (.shp) uploaded as a Table Asset.
Polarization Channel	Target Band: Selected co-polarized VV channel. Output Band Name: Sigma0_VV generated via the local process tree.	Target Band: Filtered via listContains and extracted using .select('VV').
Radiometric Calibration & dB Conversion	Operator: Calibration. Parameters: Checked Output: Sigma0 box; <i>did not</i> check the internal "Output image in dB" conversion switch. Intermediate State: Linear scaling	Operator: Ingested natively via backend cloud automated script routines. Parameters: Pre-computed and stored as log-scaled backscatter coefficient values in decibels (sigma0 in dB).

	(Sigma0_VV). Pre-processing conversion to decibels (_db) was handled downstream prior to Band Math.	
Speckle Filtering	Filter Type: Lee Sigma (Adaptive directional). Parameters (Defaults): Window Size = 7x7; Target Window Size = 3x3; Sigma = 0.9; Number of Looks = 1.	Custom code needed to match SNAP's LeeSigma adaptive edge filtering. <code>kernel = ee.Kernel.square(3, 'pixels'); var noiseVariance = 0.25;</code> <code>weight=localVariance.subtract(localMean.pow(2).multiply(noiseVariance)).divide(localVariance);weight=weight.clamp(0, 1);</code> Filter Type: Focal Median (Non-adaptive rank-order kernel spatial smoothing). Parameters: Radius = 30 meters; Kernel Shape = 'circle' also produced close results.
Terrain Correction	Operator: Range Doppler Terrain Correction. Elevation Source: SRTM 1Sec HGT (Auto-Download).	Operator: Pre-computed natively on ingestion by Google using the automated Sentinel-1 Toolbox backend.
Inundation Thresholding Logic	Operator: Band Math. Expression: if Sigma_VV_M_db < -12 then 1 else 0 executed independently across both master (August) and slave (July) bands.	Operator: applied similar Relational and conditional Boolean array steps. Expression: waterThreshold.lt(-12) applied to both target layers, followed by logical subtraction parsing: <code>postWater.and(preWater.not())</code> .
Quantitative Area Calculation Method	Tool: Analysis > Statistics computation panel. Calculated Value: 85.86 km² . Metric Space: Flat, projected Cartesian grid measurements (e.g., UTM Zone 37N square meters).	Tool: <code>ee.Reducer.sum()</code> running over a generated <code>ee.Image.pixelArea()</code> map layer. Calculated Value: 87.09 km² . Can be Optimized to 84.4 km² by tuning speckle filter at cost of loosing edges. Metric Space: GEE default assumes the curved WGS84 ellipsoid. We used <code>crs: 'EPSG:32637'</code> , // to Enforce GEE to local UTM Zone 37N Planar Cartesian projection

7.2.1 Optimization history

- var noiseVariance = 0.04; resulted 112.3345852761005 kmsq
- var noiseVariance = 0.25; resulted 87.09379531558267 kmsq
- var noiseVariance = 0.5; resulted 85.24767501784169 kmsq
- var noiseVariance = 1; resulted 84.44188657308364 kmsq

7.3 Drivers of Discrepancy between the Two Platforms

To establish which platform provides the authoritative baseline for official reporting, the overall result and the 1.23 square kilometer discrepancy must be further evaluated through the lens of radar processing mechanics. Remote sensing literature establishes that ESA SNAP provides the more mathematically rigorous and spatially accurate output for localized boundary delineation, whereas GEE introduces a predictable overestimation error.

The potential drivers behind this variance are discussed below, ranked from highest to lowest impact mainly based on the workflow assessment, Sensitivity Testing and optimization and literature reviews. However, readers should consider this as initial insight to potential drivers and we recommend to do further review on these aspects.

Speckle Filtering Order and Methods

The divergence in speckle filtering methodologies represents the primary reason the GEE output is larger than the SNAP baseline.

- **SNAP Processing (Authoritative Baseline):** In SNAP, speckle filtering is applied before terrain correction. Filtering the raw radar geometry prevents the spatial interpolation and resampling of noisy pixels. Furthermore, SNAP utilizes advanced, edge-preserving adaptive filters, such as the Refined Lee filter, which dynamically adjust search window shapes along sharp land-water boundaries. This preserves the true backscatter signal and prevents boundary bleeding.
- **GEE Processing:** Conversely, GEE hosts preprocessed GRD data that has already undergone terrain correction. Speckle filtering cannot be applied to raw geometry within GEE. The analyst must apply filtering directly to already terrain-corrected, geocoded imagery. Furthermore, GEE offers simple neighborhood reductions (such as the focal_median, Focal_mean filters) and replicating advanced adaptive filters used in SNAP (e.g. LeeSigma, Refined Lee) requires the user to construct custom scripts which are programmatically complex. Applying a rigid, non-adaptive spatial filter to resampled pixels blurs sharp land-water gradients, causing pixel-level overestimation along the winding banks of the Awash River.

Coordinate Geometries and Area Calculation

The mathematical approach to calculating surface area can also introduce minor, secondary variations between the two platforms.

- **SNAP Cartesian Approach:** By default, SNAP calculates area using a flat, projected coordinate system. It counts the total number of water-classified pixels within the UTM Zone 37N grid and multiplies that count by the nominal pixel resolution (100 square meters for 10-meter pixels).
- **GEE Geodesic Default:** GEE standard area tool, `ee.Image.pixelArea()`, calculates the true physical area of each individual pixel geodetically on the WGS84 ellipsoid. This accounts for map projection distortions, as pixel sizes vary based on their distance from the Equator. While geodesic calculations are mathematically superior for continental scales, they introduce minor discrepancies when compared to flat Cartesian baselines.

Resampling and Edge Anomalies

The lowest impact variance stems from how each platform handles raster projections and multi-temporal alignment.

- **SNAP Configuration:** SNAP utilizes explicit, user-defined resampling configurations, such as Bilinear Interpolation, during its multi-temporal collocation phase. This keeps pixel boundaries rigid and predictable.
- **GEE Configuration:** GEE scales raster projections dynamically based on the display zoom level and the specific scale arguments passed during reduction or export operations. This dynamic scaling can cause minor pixel-shifting anomalies along the outermost edges of the study area.

7.4 Validation

To evaluate the reliability of a radar-derived binary flood map, a formal validation process must be performed. Because Thresholding relies on isolating low-backscatter values from specular reflection, accuracy assessments help identify misclassified pixels caused by surface roughness variations or environmental noise.

7.4.1 Empirical Reference Data Integration

Validating the final flood map requires comparing the classified binary layers against independent ground-truth reference data collected close to the crisis date of August 7, 2025. Common reference sources for this study site include:

- **Field Observations and Crowd-Sourced Data:** Ground-truth GPS coordinates, photographs, and emergency response records provided by local authorities or field teams in the Oromia region.
- **Hydrological Gauge Data:** Correlating the spatial extent of the flood map with historical peak discharge and water level records from local river gauge stations along the Awash River.
- **High-Resolution Optical Imagery:** Utilizing cloud-free or partially clear optical data from sensors like Sentinel-2, Planet Scope, or high-resolution Google Earth updates to visually confirm inundated areas.

7.4.2 Systemic Sources of Error in C-Band SAR Flood Mapping

When validating the map against reference datasets, the technical differences between SNAP and GEE directly influence the distribution of systematic error metrics:

- **False Positives (Overestimation):** Smooth artificial surfaces like airport runways, asphalt roads, or dry, flat sandy soils can mimic the specular reflection of calm water, forcing backscatter values below the -12 dB threshold. Because GEE utilizes a post-terrain correction focal median filter, it tends to blur sharp land-water boundaries, amplifying

this overestimation error along infrastructure margins. Consequently, the SNAP workflow yields tighter, more accurate boundary delineation due to its edge-preserving Refined Lee filter.

- False Negatives (Underestimation): Flooding that occurs beneath dense forest or crop canopies often goes undetected in C-band SAR imagery due to double-bounce scattering, which makes the surface appear bright instead of dark. Wind-induced waves on open water can also increase surface roughness, raising backscatter and masking the presence of water. Both SNAP and GEE maps suffer equally from this physical limitation.

8 Operational Efficiency and Performance Trade-Offs

The operational contrast between SNAP and GEE represents a classic trade-off between interactive graphical precision and scale-invariant automation.

ESA SNAP	Google Earth Engine(GEE)
High Desktop Interaction	High Cloud Automation
Heavy Local Compute (RAM/Disk)	Server-Side Processing
Ideal for Single-Scene Tuning	Ideal for Time-Series Scaling
Manual Data Management	Immediate Programmatic Access
Less programing skill required	High programing skill required

Table 2: SNAP Vs. GEE Operational comparison

- ESA SNAP Advantages and Bottlenecks: SNAP shines in an educational setting because its Graphical User Interface (GUI) forces the analyst to explicitly understand every step of the SAR processing chain—from orbit adjustments to terrain transformations. However, it is severely bottlenecked by local system hardware limitations (CPU, RAM, and storage). Running large batch graphs or stitching multiple Sentinel-1 swaths together frequently results in out-of-memory errors on standard laboratory computers.
- GEE Advantages and Bottlenecks: GEE eliminates local hardware limitations by executing JavaScript code parallelly across Google’s distributed server infrastructure. Complex processing chains that would take hours to run locally on SNAP complete in seconds on GEE. The platform is built for rapid, automated time-series monitoring. The main drawback is its abstracted, "black-box" preprocessing environment, which hides underlying geometric corrections from the analyst and limits custom parameters.

9 Conclusion

This practical work successfully developed and compared an end-to-end processing workflow to map the severe August 2025 flood event along the Awash River plain in the Oromia region of Ethiopia. By applying a bi-temporal change detection strategy, the core physics of SAR backscatter dynamics were effectively leveraged to isolate temporary

floodwaters from permanent water channels. The smooth surface of open water induced specular reflection, causing a distinct drop in the co-polarized (VV) backscatter coefficient. Applying an empirical threshold of -12 dB successfully extracted the flood boundaries, producing highly congruent binary inundation maps across both environments.

Methodologically, the exercise provided a valuable comparative evaluation of desktop-based and cloud-based remote sensing frameworks. The ESA SNAP environment provided excellent, educational, and granular control over the step-by-step radar processing chain, yielding a highly accurate, localized baseline. Conversely, Google Earth Engine demonstrated superior operational efficiency by removing computational bottlenecks, eliminating local storage constraints, and allowing the workflow to scale seamlessly.

For final Master's research projects, these two environments are highly complementary. SNAP serves as the necessary scientific engine for phase-preserved micro-scale diagnostics (such as InSAR and polarimetric coherence). Meanwhile, GEE acts as a powerful operational engine for cloud-parallel, regional-scale environmental monitoring. Mastering both platforms provides researchers with a robust, multi-platform toolkit to address complex hydrological hazards and support climate resilience efforts.

10 References

1. Selmi et al. Flood mapping using the Sentinel-1 imagery and the ESA SNAP S1 Toolbox
2. ESA - Sentinel 1 - ESA's Sentinel Toolboxes <http://step.esa.int/main/doc/tutorials>
3. Awash river 2025 flooding news relief web <https://addisstandard.com/awash-river-flooding-displaces-thousands-in-west-and-southwest-shewa/>
4. Meyer et al. The SAR Handbook, Chapter 2 - Spaceborne Synthetic Aperture Radar: Principles, Data Access, and Basic Processing Techniques.